



Parkinson's Disease

What is Parkinson's disease?

Parkinson's disease (PD) is a disorder of the nervous system that affects movement and gets worse over time. It causes nerve cells in parts of the brain to weaken, become damaged, and die. This can lead to symptoms that include:

- Problems with movement and balance
- Uncontrollable shaking of the hands or fingers ([tremors](https://ninds.nih.gov/node/825) (<https://ninds.nih.gov/node/825>))
- Stiffness

As symptoms get worse, people with PD may have difficulty walking, talking, or completing everyday tasks.

PD and the brain

Most PD symptoms happen when brain cells weaken, become damaged, and die in a brain area called the substantia nigra. The neurons in this area produce dopamine. Dopamine is a chemical messenger that sends signals in the brain to produce smooth, purposeful movements. Research shows that by the time PD symptoms appear, most people have lost 60 to 80% or more of the dopamine-producing cells in the substantia nigra.

People with PD also have damage to brain cells that make norepinephrine. This is the main chemical messenger that's involved in things the body does on its own, like keeping the heart beating and blood pressure steady. The loss of norepinephrine in people with PD is also thought to explain some symptoms that aren't related to movement, such as being tired and blood pressure changes.

Damaged brain cells in most people with PD have Lewy bodies, abnormal clumps of a protein called alpha-synuclein. Researchers don't yet know why Lewy bodies form or how they affect PD.

[Learn more about how the brain works](https://ninds.nih.gov/node/8168) (<https://ninds.nih.gov/node/8168>).

Symptoms of PD

PD affects people in different ways, including when symptoms appear and how fast they get worse. Although PD symptoms often begin on one side of the body, usually both sides are affected. Symptoms are often less severe on one side than they are on the other. Four common symptoms of PD are:

1. **Tremors (shaking):** Often begins in a person's hand, although sometimes the person's foot or jaw can also be affected first. PD tremors have a rhythmic back-and-forth motion. Often, a tremor will cause a person to rub their thumb and forefinger together, which may look like "pill rolling." It's most obvious when the hand is at rest or when a person is under stress. This tremor usually disappears during sleep and may improve when a person moves on purpose. Some people with PD may never develop a tremor.
2. **Rigidity (muscle tightening):** This symptom affects most people with PD. Muscles stay tense and tight, causing achiness or stiffness and may cause trouble with movement. There are two types of rigidity that can occur in PD. One type, called "lead pipe" rigidity, feels like steady resistance when someone else tries to move a person's arm. The other type, called "cogwheel" rigidity, feels like short, jerky movements if someone else tries to move a person's arm.
3. **Bradykinesia:** The body moves more slowly, even with simple everyday actions that usually happen automatically. Daily activities—such as washing or dressing—that the person could once perform quickly and easily may take much longer. A person with PD may show fewer facial expressions (known as "masked face").
4. **Postural instability:** Balance problems and changes in posture that can increase the risk of falls.

People with PD often develop a parkinsonian gait. This includes a tendency to stoop forward with less natural swinging in one or both arms and they may take small, shuffling steps. Some people with PD may have trouble trying to move, which is called start hesitation. They may have trouble getting their legs to take a step and may look like they are freezing in place. When walking, each step may get faster and faster and a person may have a hard time stopping or slowing down. This is called festination.

People with PD may have other symptoms, such as:

Mental and emotional health problems	—
These can include depression or anxiety and may happen during the early stage of PD, even before the start of movement problems.	
Difficulty with swallowing and chewing	—
Problems with swallowing and chewing may happen in later stages of PD. Food and saliva may collect in the mouth and back of the throat, which can cause choking or drooling. It can be difficult for people in the later stages of PD to eat properly and get enough nutrients from food.	
Speech changes	—
People with PD may have speech problems. It may seem like they are speaking quietly or in a flat voice without much change in tone. Some people may hesitate before speaking. They may also slur or speak too fast.	
Urinary and digestive problems	—
People with PD may feel the need to go to the bathroom a lot, including at night, or have difficulty emptying their bladder. Digestive problems are also common in PD and often develop over time. Changes in the nerves that control the gut can affect swallowing and digestion, leading to symptoms like nausea and constipation—and in some cases, more serious complications.	
Skin problems	—
People with PD may have oily skin, especially on the forehead and at the sides of the nose. The scalp may also become oily, which can cause dandruff. In other cases, a person with PD may have very dry skin or sweat a lot.	
Sleep problems	—
Common sleep problems in PD include difficulty staying asleep, tossing and turning, and nightmares or very emotional dreams. People may also feel drowsy or suddenly fall asleep during the day. Some people with PD act out their dreams (which is called REM sleep behavior disorder). People with these symptoms have changes in thinking and behavior that cause the condition to get worse quickly.	
Dementia or other cognitive problems	—
Some people with PD develop problems thinking and remembering things. These types of problems can get worse over time, and some people may develop Parkinson's disease dementia (https://ninds.nih.gov/node/632/#toc-dementias-associated-with-aging-and-neurodegeneration). PD can also affect a person’s ability to judge how deep or how far away things are. They may also have trouble paying attention, speaking, and figuring things out.	
Orthostatic hypotension	—
This happens when a person’s blood pressure drops very quickly when they stand up after sitting or lying down. This symptom may cause dizziness and lightheadedness. In extreme cases, a person may lose balance or faint.	
Muscle cramps and dystonia	—
Muscle stiffness and the inability to move normally in people with PD often causes muscle cramps, especially in the legs and toes. PD can also lead to dystonia (https://ninds.nih.gov/node/10787)—when muscles tighten for a long time, causing the body to twist or hold awkward positions.	

Pain	—
People with PD may have aches and pains in their muscles and joints because of muscle tightening and unusual body positions that happen with the disorder.	
Fatigue and loss of energy	—
Many people with PD are very tired, especially late in the day. This fatigue can be due to depression or sleep problems, but it can also be related to muscle problems—such as difficulty with moving, tremors, or stiffness. Some medications used to treat PD may also cause tiredness.	
Sexual problems	—
PD can affect sexual function because it changes how the brain and nerves control the body. PD-related depression or use of certain medicines may also lower sex drive.	

PD is the most common form of parkinsonism, a group of disorders with symptoms like Parkinson's disease. Many disorders can cause symptoms similar to those of PD, including:

- [Multiple system atrophy](https://ninds.nih.gov/node/730)
- [Lewy body dementia](https://ninds.nih.gov/node/633)
- [Progressive supranuclear palsy](https://ninds.nih.gov/node/779)
- [Corticobasal degeneration](https://ninds.nih.gov/node/626)

These disorders are sometimes referred to as “atypical Parkinsonisms” or “Parkinson's-plus syndrome” because they have other symptoms in addition to PD symptoms. They often don’t get better with the medicine levodopa.

In very rare cases, parkinsonian symptoms may happen in people younger than 20. This condition, juvenile parkinsonism, often starts with muscle tightening and slow movement. These symptoms often improve with the medicine levodopa.

Who is more likely to get Parkinson’s disease?

In most people with PD, the cause is unknown. Some cases of PD are inherited (passed down in families) and can be traced to specific genetic variations (mutations). But researchers think risk for PD comes from both genes **and** environmental factors. The term “exposome” refers to all environmental and lifestyle exposures across a person’s lifetime.

Non-genetic risk factors linked to PD

While it’s difficult to definitively predict who will develop PD, several factors are associated with increased risk. These include:

- **Age:** The average age for PD to start is in a person’s early to mid-60s, and the chances are higher as people get older. A small percent of people with PD have “early-onset” Parkinson’s disease, which begins before 50.
- **Biological sex:** PD affects more men than women.
- **Family history:** People with one or more close relatives who have PD are more likely to get it themselves. This could be from having the same genes or similar surroundings and daily habits.
- **Environmental exposure:** People who have been exposed to some types of pesticides and other pollutants have a greater risk for getting PD. For example, people can be exposed to pollutants through farming or landscaping.
- **Other factors:** Several additional things can affect a person’s risk for PD—including having other health conditions, gut bacteria, sleep, and stress.

Genes linked to PD

Some PD gene variations affect energy production inside cells through a cell-damaging process called oxidative stress. This happens when energy production doesn’t work properly and harmful molecules build up.

Several genes and the proteins they make are linked to PD. These include:

DJ-1	—
The <i>DJ-1</i> gene makes a protein that helps protect cells from damage caused by harmful molecules. Variations in this gene can cause rare, early-onset forms of PD, when a person gets PD before 50.	

GBA
The <i>GBA</i> gene makes the protein glucocerebrosidase-beta. Variants in the <i>GBA1</i> gene are the most common genetic risk factor for Parkinson’s disease. Variations in this gene can also cause Gaucher disease, a type of lipid storage disorder (https://ninds.nih.gov.com/node/706). Changes in the <i>GBA1</i> gene affect lysosomes, the parts of cells that break down and recycle proteins. This can cause proteins to build up and harm cells.
LRRK2
The <i>LRRK2</i> gene makes a protein that helps control several important processes in cells, including how materials move inside cells, immune system activity, and how cells recycle waste. When the LRRK2 protein does not work properly, cells may have trouble clearing extra alpha-synuclein. Variations in the <i>LRRK2</i> gene may lead to the clumping of Lewy bodies. Researchers think the LRRK2 protein may be unusually active in people with PD who do not have an inherited form of the disorder.
PINK1
The <i>PINK1</i> gene makes a protein that detects damage to a cell’s mitochondria. The PINK1 protein works with the protein parkin to help cells break down and remove damaged mitochondria before they build up and harm the cell. Variations in this gene cause damaged mitochondria to build up and are also linked to early-onset forms of PD.
PRKN
The <i>PRKN</i> gene makes the protein parkin that helps cells identify and remove damaged mitochondria, the parts of cells that make energy. Variations in this gene can also cause early-onset forms of PD.
SNCA
The <i>SNCA</i> gene makes the protein alpha-synuclein that helps nerves communicate with each other and was the first gene to be linked to PD. This led to the discovery that Lewy bodies in the brains of people with PD contain clumps of abnormal alpha-synuclein. This discovery also showed how inherited and non-inherited PD are similar.

How is Parkinson’s disease diagnosed and treated?

Diagnosing PD

There isn’t a specific, single test that can definitively diagnose PD. Instead, doctors consider:

- A person’s **medical history** and a **neurological examination** to look for problems with movement and other typical PD symptoms.
- **Blood and other laboratory tests** to check for other disorders that may be causing symptoms and identify biological signatures called biomarkers that help doctors confirm a diagnosis. The Syn-One Test detects variations of alpha-synuclein in small skin samples. NIH-supported research led to the development of this test. Similarly, the SAAmplify-αSYN test detects tiny amounts of alpha-synuclein in spinal fluid and was also funded in part by NIH research. These two tests can help diagnose PD but can’t tell PD from other similar disorders, such as multiple system atrophy and Lewy body dementia.
- **Brain imaging tests** such as SPECT (single-photon emission computed tomography) and MRI (magnetic resonance imaging) can rule out other disorders. A dopamine active transporter (DaT) scan can help diagnose PD, but it can’t distinguish PD from atypical Parkinsonism.

Genetic testing can identify genetic variations in inherited forms of PD to predict a person’s risk for developing the disorder. If you think someone in your family may have PD or that you might have a genetic variation linked to PD, ask a doctor about genetic testing. A genetic counselor can help explain what the results mean.

[Learn more about neurological diagnostic tests and procedures](#) (<https://ninds.nih.gov.com/node/10003>).

Treating PD

There is no cure for PD, but medicines or surgery can help people with PD move more easily and improve other symptoms. People with PD should talk to a doctor to decide which PD treatments are best for them.

There are three types of medicines used to treat PD:

- Medicines that increase brain dopamine levels. The most common ones, such as levodopa, are pills that turn into dopamine after they enter the brain. Levodopa is almost always combined with another medicine, called carbidopa, to help more of it reach the brain and to reduce the risk of nausea as a side effect. This combination is called carbidopa-levodopa.
- Medicines that change levels of other brain chemicals, such as acetylcholine, to ease PD tremors.
- Medicines that help treat PD symptoms unrelated to the ability to move. For example, people with PD-related depression may be prescribed antidepressant medicines.

Movement-related PD symptoms may get much better after first taking medicines. Over time, medicines may work for shorter periods, and their effects may be less predictable. Side effects may also develop. When recommending treatment for PD, a doctor will consider how symptoms affect a person's daily life and choose treatments based on their needs.

Since no two people with PD will react the same way to a medicine, it may take time and patience to get the dose and combination of medicines right. Even then, PD symptoms may not completely go away.

People with PD who are taking medicines need to be monitored by a doctor to look for side effects. For example, in some people with PD, certain medicines used to treat PD may cause people to develop strong urges or repeat certain behaviors. In some people, PD medicines may also cause hallucinations, dyskinesia (unwanted movements such as twisting and writhing), or beliefs that don't match reality.

Doctors usually recommend the following medicines:

Carbidopa-levodopa

People can't take dopamine by itself because dopamine doesn't easily travel from the bloodstream into the brain. This is because of the [blood-brain barrier](https://ninds.nih.gov/node/9714#blood-brain-barrier) (<https://ninds.nih.gov/node/9714#blood-brain-barrier>), a protective layer of cells inside brain blood vessels that tightly controls what substances get into the brain.

The main PD therapy is the medicine levodopa (also known as L-dopa). Levodopa can ease movement-related PD symptoms, but it doesn't replace lost nerve cells or stop the disease from getting worse. But the body quickly breaks down levodopa, so doctors combine it with another medicine (carbidopa).

Carbidopa blocks levodopa breakdown in the body, so more levodopa can get into the brain. Once in the brain, levodopa is converted to dopamine. Carbidopa-levodopa usually works very well to reduce movement-related PD symptoms during the early stages of the disorder. A doctor may need to increase the dose over time.

Carbidopa-levodopa can reduce or eliminate tremors in many people with PD. But some PD tremors don't respond to carbidopa-levodopa. A doctor may prescribe a different type of medicine to treat tremors in these people.

After taking carbidopa-levodopa, people with PD may have side effects. These include nausea, low blood pressure, restlessness, and feeling sleepy.

The longer a person has PD, symptoms may start happening in the morning before their first dose of carbidopa-levodopa and in between doses. People may experience sudden, unpredictable "off periods" where the medicine doesn't seem to be working. In these cases, taking carbidopa-levodopa more often and in smaller amounts can help.

People with PD should never stop taking carbidopa-levodopa without talking with a doctor because stopping it suddenly can have potentially serious side effects.

Dopamine agonists

Dopamine agonist medicines (apomorphine, pramipexole, ropinirole, and rotigotine) act like dopamine in the brain. They're common treatments in early-stage PD or in combination with carbidopa-levodopa in later stages of PD. A new medicine, Onapgo, is a small under-the-skin device that gives a continuous dose of apomorphine. It can help reduce "off" periods when PD symptoms return.

Some of the potential side effects with these medicines are similar to those associated with the use of carbidopa-levodopa, including:

- Drowsiness

- Suddenly falling asleep
- Hallucinations
- Confusion
- Dyskinesia
- Edema (swelling due to too much fluid in body tissues)
- Nightmares
- Vomiting

In rare cases, these medicines can cause strong urges to gamble, shop, have sex, or engage in other behaviors that are hard to control.

MAO-B inhibitors

MAO-B inhibitor medicines (selegiline, rasagiline, and safinamide) slow the breakdown of dopamine in the brain, allowing more dopamine to build up and ease PD symptoms. NIH-supported research showed that selegiline (also called deprenyl) can delay the need for carbidopa-levodopa treatment by up to a year or more.

Many people with PD find MAO-B inhibitors easy to take, although side effects may include nausea, difficulty sleeping, and a drop in blood pressure when standing up that can cause dizziness or fainting.

COMT inhibitors

COMT inhibitor medicines (entacapone, opicapone, and tolcapone) stop the breakdown of dopamine in the brain, easing PD symptoms. These medicines can make carbidopa-levodopa last longer and lessen “off periods” by increasing levels of dopamine. Side effects may include:

- Diarrhea
- Nausea
- Difficulty sleeping
- Dizziness
- Changes in urine color
- Stomach pain
- Low blood pressure
- Hallucinations

In rare cases, tolcapone can cause severe liver disease, and people taking tolcapone should have a doctor check their liver function regularly.

Anticholinergics

Anticholinergic medicines (trihexyphenidyl and benztropine) reduce levels of the brain chemical acetylcholine. In people with PD, changes in the balance between dopamine and acetylcholine affect movement and can cause symptoms like tremors. Anticholinergic medicines help restore this balance and can be particularly effective for PD tremors. Side effects may include:

- Dry mouth
- Constipation
- Trouble urinating
- Hallucinations
- Memory loss
- Blurred vision
- Confusion

Amantadine

This medicine can help with PD symptoms (tremors, stiffness, and uncontrolled movements) because it affects levels of dopamine and other brain chemicals. People can take this medicine by itself soon after symptoms develop. It’s also often combined with other PD medicines, like anticholinergics or carbidopa-levodopa. Amantadine can also help reduce involuntary movements caused by other PD medicines.

In about half of the people who take it, amantadine may stop treating some PD symptoms like stiff and slow movements after several months. However, it may continue to help reduce dyskinesia for much longer. Amantadine's side effects may include:

- Difficulty sleeping
- Patchy, colored skin
- Edema (swelling due to too much fluid in body tissues)
- Feeling restless or upset
- Hallucinations

Researchers don't know for sure, but they think amantadine increases dopamine's effects.

Other treatments for PD can include:

Deep brain stimulation

[Deep brain stimulation \(DBS\)](https://www.ninds.nih.gov/health-information/disorders/deep-brain-stimulation-dbs) (<https://www.ninds.nih.gov/health-information/disorders/deep-brain-stimulation-dbs>) is usually considered for people with PD after medicines stop working well or cause difficult side effects. DBS involves brain surgery in which a neurosurgeon places a thin wire that carries electrical signals into the part of the brain that controls movements.

Like a pacemaker in the heart, a battery pack placed in the chest area under a person's collarbone sends electrical signals to the wire placed in the brain. These signals help change the abnormal brain activity that causes PD movement problems, but DBS doesn't cure the disease.

While DBS helps with movement-related symptoms, it doesn't improve other symptoms—like speech problems, “freezing,” difficulty with body positioning and balance, anxiety, depression, or dementia.

NIH-funded researchers helped develop adaptive DBS, a newer form of DBS that automatically adjusts signals in real time. This treatment gives the brain only as much help as needed to better control PD symptoms.

Surgery

Doctors may recommend surgery for some people with PD when medicines no longer work well enough to treat their symptoms. One type of surgery for PD is lesion surgery. This surgery treats PD by carefully destroying small areas of the brain that cause symptoms.

The most common type of lesion surgery is pallidotomy because it targets a part of the brain called the globus pallidus. This brain area helps control how smooth and controlled a person's movements are. Pallidotomy can improve PD symptoms like tremors, rigidity, and bradykinesia (when the body moves too slowly). Research also suggests that pallidotomy can help with walking and balance. It may also allow people with PD to take lower doses of carbidopa-levodopa.

Another type of surgery for people with PD is thalamotomy because this surgery targets a part of the brain called the thalamus. This part of the brain acts like a relay center, passing and coordinating movement-related signals between brain regions.

Because these surgeries permanently destroy small amounts of brain tissue, they have mostly been replaced by DBS. Another treatment uses focused ultrasound, which uses sound waves from outside the head to treat symptoms without traditional surgery.

Lifestyle changes and complementary and supportive therapies

Lifestyle changes may help with PD, including:

- **Exercise:** Physical activity—such as walking, dancing, yoga, cycling, swimming, and other activities—can help people with PD. Exercise can improve flexibility, balance, strength, and the ability to move. People with PD should always check with a doctor before beginning a new exercise program. NIH-supported researchers are conducting clinical trials to test whether exercise that makes a person breathe hard and sweat can help keep PD from getting worse.
- **A healthy diet:** A healthy diet can help ease some PD symptoms. Eating a fiber-rich diet and drinking plenty of fluids can help with constipation. Too much protein may limit a person's ability to absorb PD medicines like carbidopa-levodopa. NIH is funding research to determine if caffeine, dietary supplements, and other foods or nutrients may help prevent or treat PD.

Some people with PD use complementary and supportive non-medicine therapies to help ease PD symptoms:

- Physical, occupational, and speech therapies to help with walking and speech, tremors and rigidity, and coping with memory and thinking changes
- Voice exercises to improve speech and swallowing
- Hypnosis
- Massage therapy
- [Acupuncture](https://www.nccih.nih.gov/health/acupuncture-what-you-need-to-know) (<https://www.nccih.nih.gov/health/acupuncture-what-you-need-to-know>)

Living with PD

Even though PD usually takes a long time to get worse, a person's day-to-day life will change in many ways that affect work and relationships. These changes can be hard to accept.

Support groups can help people learn how to live with PD's effects and offer valuable information, advice, and experience. Support groups made up of people who have some experience with the disease can help people with PD, their families, and caregivers deal with many issues. These include finding doctors familiar with PD and living with both physical limitations and effects on mood. Individual or family counseling may also help people find ways to live as well as they can with PD.

Many people with PD continue to work either full- or part-time, although they may need to adjust their schedule and working environment to meet their personal needs. In the late stages of PD, medicines may no longer control symptoms well. Serious effects on the body may include choking, pneumonia, and falls.

What are the latest updates on Parkinson's disease?

The National Institutes of Health (NIH (<https://ninds.nih.gov.com/glossary?title=NIH>)), which includes NINDS (<https://ninds.nih.gov.com/glossary?title=NINDS>), is the leading federal funder of research on the brain and nervous system, including disorders such as PD. NIH supports new and innovative research to better understand, diagnose, and treat PD. NINDS is the main NIH institute supporting PD research. Learn more about [NINDS's research focus on Parkinson's disease](https://ninds.nih.gov.com/node/19) (<https://ninds.nih.gov.com/node/19>).

Mostly through NINDS (<https://ninds.nih.gov.com/glossary?title=NINDS>), NIH (<https://ninds.nih.gov.com/glossary?title=NIH>) has key programs and resources for PD research, which include:

- The [NINDS Intramural Research Program](https://ninds.nih.gov.com/node/10507) (<https://ninds.nih.gov.com/node/10507>) conducts clinical research to understand how PD affects the body. This information may lead to better treatments.
- The [National Plan to End Parkinson's Act](https://www.ninds.nih.gov/current-research/trans-agency-activities/national-plan-end-parkinsons) (<https://www.ninds.nih.gov/current-research/trans-agency-activities/national-plan-end-parkinsons>) is creating a national strategy to improve Parkinson's disease research, care, and prevention. The plan brings together scientists, doctors, and patient groups to help speed progress toward better treatments.
- The [NINDS Morris K. Udall Centers of Excellence for Parkinson's Disease Research program](https://ninds.nih.gov.com/node/2267) (<https://ninds.nih.gov.com/node/2267>) supports research centers across the country that work together to study how PD affects the body, altered gene patterns in PD, and new treatment strategies.
- [The Parkinson's Disease Biomarkers Program](https://pdpb.ninds.nih.gov/) (<https://pdpb.ninds.nih.gov/>) is working to discover ways to identify people at risk for developing PD and to track the disease over time. It funds research and collects human biological samples and clinical data to identify biomarkers (biological signs of disease that can be found and measured) that will help researchers find new PD treatments faster. This program is also studying disorders that are biologically similar to PD like progressive supranuclear palsy, multiple system atrophy, and Lewy body dementia (including Parkinson's disease dementia).
- The [Accelerating Medicines Partnership® for Parkinson's Disease and Related Disorders](https://amp-pd.org/) (<https://amp-pd.org/>) program involves NIH, biopharmaceutical companies, and nonprofit organizations. It allows scientists to use shared data from many clinical studies to try to understand these diseases. The program speeds up research and identifies ways to develop new treatments for people with PD and related disorders or those at risk.
- [NINDS Biospecimen Exchange for Neurological Disorders](https://biosend.org/) (<https://biosend.org/>) collects tissue samples and body fluids (like blood and cerebrospinal fluid) from NINDS-supported research, including phase 2 and 3 clinical trials. These samples are shared with researchers to find biomarkers for PD and related disorders.
- [NINDS Human Cell and Data Repository](https://nindsgenetics.org/) (<https://nindsgenetics.org/>) provides cells to researchers to help them study neurological disorders, including PD.
- [NIH NeuroBioBank](https://neurobiobank.nih.gov/) (<https://neurobiobank.nih.gov/>) safely collects and stores human brain tissue samples collected after death and related tissue samples that are crucial to understanding healthy brain function and nervous system disorders, including PD. Learn more about [brain donation](https://ninds.nih.gov.com/node/9736) (<https://ninds.nih.gov.com/node/9736>).

For more information on research about PD, check [NIH RePORTER](https://reporter.nih.gov/) (<https://reporter.nih.gov/>), a searchable database of current and past research projects funded by NIH (<https://ninds.nih.gov.com/glossary?title=NIH>) and other federal agencies. RePORTER also has links to publications and resources from these projects.

For research articles and summaries on PD, search [PubMed](https://pubmed.ncbi.nlm.nih.gov/) (<https://pubmed.ncbi.nlm.nih.gov/>), which contains citations from medical journals and other sites.



Learn About Clinical Trials

Clinical trials are studies that allow us to learn more about disorders and improve care. They can help connect patients with new and upcoming treatment options.

Search Clinical Trials<https://clinicaltrials.gov/ct2/home>

How can I or a loved one help improve care for people with Parkinson's disease?

Consider participating in a clinical trial, so clinicians and scientists can learn more about PD and related disorders. Clinical research with human study participants helps researchers learn more about a disorder and perhaps find better ways to detect, treat, or prevent disease safely.

All types of participants are needed—those who are healthy or may have an illness or disease—of all different ages, sexes, races, and ethnicities. This makes sure that study results apply to as many people as possible and that treatments will be safe and effective for everyone who will use them.

For information about participating in clinical research, visit the [NINDS Clinical Trials site](https://ninds.nih.gov.com/health-information/clinical-trials-and-studies) (<https://ninds.nih.gov.com/health-information/clinical-trials-and-studies>) and [NIH Clinical Research Trials and You](https://www.nih.gov/health-information/nih-clinical-research-trials-you) (<https://www.nih.gov/health-information/nih-clinical-research-trials-you>). Learn about clinical trials currently looking for people with PD at [ClinicalTrials.gov](http://clinicaltrials.gov/) (<http://clinicaltrials.gov/>), a searchable database of current and past clinical studies and research results.

Where can I find more information about Parkinson's disease?

Information may be available from the following sources:

- [American Parkinson Disease Association](http://www.apdaparkinson.org/) (<http://www.apdaparkinson.org/>)
800-223-2732
apda@apdaparkinson.org (<mailto:apda@apdaparkinson.org>)
- [Cure MAPT FTD](https://gcc02.safelinks.protection.outlook.com/?url=https%3A%2F%2Fwww.curemaptftd.org%2F&data=05%7C02%7Cgarnetts%40ninds.nih.gov%7C2e16e00062b2433257b708dcc69856d5%7C14b77578977342d58507251ca2dc2b06%7C0%7C0%7C638603604801593733%7CUnknown%7CTWFpbGZsb3d8eyJWljojMC4wLjAwMDAiLCJQIjoiV2luMzliLCJBTiI6IjEhaWwiLCJXVCi6Mn0%3D%7C0%7C%7C%7C&sdata=Rnhn6HphoqMxPi27nGjDxcnX20GGVhq9K2gYkVcOrs%3D&reserved=0) (<https://gcc02.safelinks.protection.outlook.com/?url=https%3A%2F%2Fwww.curemaptftd.org%2F&data=05%7C02%7Cgarnetts%40ninds.nih.gov%7C2e16e00062b2433257b708dcc69856d5%7C14b77578977342d58507251ca2dc2b06%7C0%7C0%7C638603604801593733%7CUnknown%7CTWFpbGZsb3d8eyJWljojMC4wLjAwMDAiLCJQIjoiV2luMzliLCJBTiI6IjEhaWwiLCJXVCi6Mn0%3D%7C0%7C%7C%7C&sdata=Rnhn6HphoqMxPi27nGjDxcnX20GGVhq9K2gYkVcOrs%3D&reserved=0>)
hello@curemaptftd.org (<mailto:hello@curemaptftd.org>)
- [Davis Phinney Foundation](http://www.davisphinneyfoundation.org/) (<http://www.davisphinneyfoundation.org/>)
866-358-0285
contact@dpf.org (<mailto:contact@dpf.org>)
- [Lewy Body Dementia Association](http://www.lbda.org/) (<http://www.lbda.org/>)
800-539-9767
support@lbda.org (<mailto:support@lbda.org>)
- [Michael J. Fox Foundation for Parkinson's Research](http://www.michaeljfox.org/) (<http://www.michaeljfox.org/>)
212-509-0995
info@michaeljfox.org (<mailto:info@michaeljfox.org>)
- [Northwest Parkinson's Foundation](http://www.nwpcf.org/) (<http://www.nwpcf.org/>)
877-980-7500
info@nwpcf.org (<mailto:info@nwpcf.org>)

[Parkinson's Foundation](https://parkinson.org/) (https://parkinson.org/)
800-473-4636
helpline@Parkison.org (mailto:helpline@Parkison.org)

[Parkinson & Movement Disorder Alliance](https://www.pmdalliance.org/) (https://www.pmdalliance.org/)
800-256-0966

[Parkinson's Resource Organization](http://www.parkinsonsresource.org/) (http://www.parkinsonsresource.org/)
877-775-4111
info@parkinsonsresource.org (mailto:info@parkinsonsresource.org)

[Parkinson Voice Project](https://parkinsonvoiceproject.org/) (https://parkinsonvoiceproject.org/)
469-375-6500
contact@parkinsonvoiceproject.org (mailto:contact@parkinsonvoiceproject.org)

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- [Deep Brain Stimulation \(DBS\)](https://ninds.nih.gov/node/631) (https://ninds.nih.gov/node/631)
- [Dementias](https://ninds.nih.gov/node/632) (https://ninds.nih.gov/node/632)
- [Lewy Body Dementia](https://ninds.nih.gov/node/633) (https://ninds.nih.gov/node/633)
- [Multiple System Atrophy](https://ninds.nih.gov/node/730) (https://ninds.nih.gov/node/730)
- [Progressive Supranuclear Palsy \(PSP\)](https://ninds.nih.gov/node/779) (https://ninds.nih.gov/node/779)
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